Joshua Fonseca Rivera

AI welfare & safety researcher | LLM introspection & character training

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Anthropic Fellow working on model welfare and character training. First-author research on LLM introspection (COLM 2026) and psychometric safety evaluation. Investigating how training may support AI flourishing, with a background in mechanistic interpretability and production engineering.

Research appointments

Anthropic Fellows Program 2026–Present
AI Safety Research Fellow | Model welfare & character training Berkeley, CA

Independent Research 2025–2026
Led *Steering Awareness*; collaborated on empirical ML research with UK AI Security Institute researchers.

Selected publications & preprints

Steering Awareness: Detecting Activation Steering from Within
Joshua Fonseca Rivera, David Demitri Africa
Conference on Language Modeling (COLM), 2026 | arXiv:2511.21399

- Fine-tuned seven instruction-tuned models (7B–70B) to detect injected activation vectors; the best achieved **95.5% detection** and **71.2% concept identification** on held-out concepts, with no false positives on clean evaluation inputs.
- Used residual-stream tracking, attention-head ablations, and causal interventions to identify a distributed transformation into a shared detection direction.
- Tested noise, misleading prompts, and novel vector constructions; established geometry-dependent transfer and increased steering susceptibility after detection training.

Measuring Activation Control in Large Language Models
Marek Mateusz Kowalski, **Joshua Fonseca Rivera**, Uzay Macar, David Demitri Africa
Preprint, August 2026 | arXiv:2608.21664 | Code | Dataset

- Co-developed the Activation Controllability Benchmark: six measures of control over residual-stream representations, evaluated across **25 models from seven families** (4B–744B).
- Found partial evasion of four monitor families on simple tasks, including linear probes and activation oracles; characterized coarse temporal control and failure of layer targeting.

Item Response Theory for AI Safety
Joshua Fonseca Rivera, Neil Shah, David Demitri Africa, Konstantinos Voudouris
Preprint, August 2026 | arXiv:2608.05086

- First-author study applying psychometrics to **eight safety benchmarks across 192 language models**; identified distinct dimensions of refusal strictness, truthfulness, and contextual harm.
- Recovered several individual benchmark scores with roughly ten adaptively selected items, reducing evaluation cost by **97–99%**; evaluated audits for prompted sandbagging and model substitutions behind APIs.

Touch&Fold: A Foldable Haptic Actuator for Rendering Touch in Mixed Reality
Shan-Yuan Teng, Pengyu Li, Romain Nith, **Joshua Fonseca**, Pedro Lopes
ACM CHI, 2021 | **Best Paper Honorable Mention (top 5%)** | Paper

Education

The University of Texas at Austin M.S., Artificial Intelligence 2026

The University of Chicago B.S., Computer Science 2020
Dean's List, 2016–2020.

Research writing & invited talks

What Happens When a Model Thinks It Is AGI?
Joshua Fonseca Rivera, David Demitri Africa | Research blog, April 2026 | Write-up | Code & data

- Extended prior consciousness self-claim experiments to AGI/ASI identities; fine-tuned GPT-4.1, Qwen3-30B, and DeepSeek-V3.1 and evaluated their behavior in multi-turn Petri scenarios and Anthropic's Agentic Misalignment evaluations.
- Built a reproducible training and evaluation pipeline with an Inspect AI provider for Tinker and action-based scoring rubrics. Reported behavioral shifts in GPT-4.1 and mixed results across the other models, with matched controls and explicit study limitations.

Invited talk: *Steering Awareness: Detecting Activation Steering from Within*, BlueDot Impact, 2026.

Engineering experience

Taekus Jan 2024–2026
Software Engineer, then Lead Mobile Engineer Remote

- Led development of a production banking app spanning **150+ screens** using React Native, Expo, and TypeScript; integrated native Swift/Kotlin wallet provisioning.
- Reduced login **p95 latency from over 15 seconds to under 2 seconds** by parallelizing authentication requests.
- Shipped check deposit and mailing from inception; rebuilt statement credits as idempotent, self-healing workflows with reconciliation, eliminating double-credit incidents and manual recovery.

Root16 Jul 2020–Apr 2023
Software Engineer Remote

- Led C#/.NET delivery for clients including Energizer, RSM, and Plante Moran; built data-migration tooling and Azure CI/CD pipelines, and mentored junior engineers.

Earlier research & teaching

Human–Computer Integration Lab, University of Chicago Jan–Jun 2020
Research Assistant | Advisor: Prof. Pedro Lopes Chicago, IL

- Built VR/AR demonstrations in Unity with Leap Motion and ran haptics user studies; contributed to *Touch&Fold* (CHI 2021, Best Paper Honorable Mention).

Teaching Assistant, Emerging Technologies Winter 2020
University of Chicago.

Technical skills

ML & experimentation: Python, PyTorch, Hugging Face Transformers, LoRA fine-tuning, activation steering, residual-stream analysis, causal interventions, model evaluation.

Evaluation & measurement: Petri, Inspect AI, Tinker, item response theory, adaptive testing, behavioral controls, benchmark validation.

Engineering: TypeScript, React Native, Django, PostgreSQL, AWS, C#/.NET, Swift/Kotlin.

Selected open-source project: Animal Crossing LLM mod. Integrated LLM dialogue into a GameCube game through live memory mapping and Python IPC.